Eric Wentz Part 2 Report: Track B — Detection & Tracking

Methodology

Track B required detecting all players across a short basketball clip, maintaining consistent track IDs through occlusions, and producing MOT-format output. The implementation draws entirely on open-source tooling, which was a deliberate choice: the interesting work is in how the pieces connect, not in rebuilding what already exists.

Detection uses Roboflow's pretrained basketball model (`basketball-player-detection-3-ycjdo/4`), built on RF-DETR and fine-tuned on basketball footage. Each frame is decoded using the supervision library, passed through RF-DETR to produce per-frame bounding boxes, and filtered to the player class. Those detections are then fed into ByteTrack, which associates them across frames via IoU matching and assigns persistent track IDs. The final output is both an annotated video and a MOT-format text file, one row per detection per frame with columns for frame number, track ID, bounding box coordinates, and confidence score (`frame, track_id, x, y, w, h, conf, -1, -1, -1`). The pipeline runs end-to-end on a Colab T4 GPU with model weights pulled automatically via the Roboflow inference API. No fine-tuning was performed on the evaluation data.

Evaluation Set

We used the SportsMOT dataset (Cui et al., 2023), a public multi-object tracking benchmark for sports footage. The specific sequence used was `v_-6Os86HzwCs_c001` from the train split, downloaded via HuggingFace (`MCG-NJU/SportsMOT`). The clip is 825 frames, approximately 33 seconds, with 8,074 ground truth detections already in MOT format, matching our output format directly.

The key limitation is an intentional domain gap: our RF-DETR model was fine-tuned on NBA broadcast footage, while this SportsMOT sequence appears to be women's overseas basketball, with different jersey styles, court markings, and camera angles. This stress-tests generalization rather than measuring best-case performance, which is a more honest evaluation of how the pipeline would behave in the wild.

Quantitative Results

We report MOTA (Multiple Object Tracking Accuracy), computed as one minus the sum of false positives, false negatives, and identity switches divided by total ground truth detections (Bernardin & Stiefelhagen, 2008).A MOTA of 0.603 is a reasonable baseline given the domain gap. HOTA would be the preferred metric in a production setting as it weights detection and association quality equally (Luiten et al., 2021), but requires a full TrackEval setup and was out of scope for this implementation.

Qualitative Examples

The annotated output video is in `results/macalester_short_annotated.mp4` and shows consistent track ID labels across frames. Selected screenshots in `results/` illustrate three cases: a clean tracking frame with all players correctly identified, an occlusion frame where players cross paths, and a failure case where a sideline figure is picked up as a player. Other types of failure cases and successes are also documented.

Error Analysis

The high false positive count (1,665) is the most significant failure. The model was trained on NBA footage where the visual context around the court is distinct from overseas basketball: different sideline layouts, jersey styles, and crowd proximity all contribute to fans, coaches, and referees being picked up as players. A court region mask filtering detections outside the playing area would eliminate a significant portion of these without any retraining.

Identity switches (423) are a structural limitation of ByteTrack's IoU-only matching. When players occlude each other or cross paths, ByteTrack has no appearance features to distinguish them after the occlusion resolves, so IDs swap. The fixed camera setup helps, since there is no camera motion to compensate for, but it does not solve the re-identification problem after contact.

The domain gap between NBA training data and overseas evaluation footage underlies both failure modes. Different jersey colors, court markings, and camera angles all push the model outside its training distribution, degrading both detection confidence and tracking stability.

Metric	Value
MOTA	0.603
False Positives	1,665
False Negatives	1,117
Identity Switches	423
Ground Truth Detections	8,074

Next Steps

With two more weeks, there are several high-value improvements across both the detection and tracking sides of the pipeline.

On the tracking side, replacing ByteTrack with BoT-SORT or SAM2 would be the first move. BoT-SORT adds appearance embeddings and camera motion compensation on top of IoU matching, which would directly address the identity switch problem when players cross paths or occlude each other. SAM2 would go further, tracking pixel-level segmentation masks rather than bounding boxes, giving the system a much more robust handle on player identity through contact and occlusion. The tradeoff is compute: SAM2 requires a GPU and would push inference time up significantly, but for a batch processing use case that is acceptable.

On the detection side, fine-tuning RF-DETR on in-domain footage matching the target league, camera angle, and jersey style would reduce both false positives and missed detections. The domain gap between NBA training data and overseas evaluation footage was the single biggest driver of our error rates, and more representative training data is the cleanest fix. Adding a court region mask to filter detections outside the playing area would also eliminate a large portion of sideline false positives cheaply, without any retraining required.

Finally, setting up full TrackEval to compute HOTA alongside MOTA would give a more complete and honest picture of tracking quality, separating detection accuracy from association accuracy in a way MOTA alone cannot. This is the evaluation upgrade that would matter most for any production deployment.

AI Usage

Claude Code (Anthropic) was used throughout for code structure, implementation, debugging, and research on model selection. All model choices, evaluation design, analysis, and writeup are the author's own.

References

Bernardin, K., & Stiefelhagen, R. (2008). Evaluating multiple object tracking performance: The CLEAR MOT metrics. *EURASIP Journal on Image and Video Processing*, 2008, 1–10.

Cui, Y., Zeng, C., Zhao, X., Yang, Y., Wu, G., & Wang, L. (2023). SportsMOT: A large multi-object tracking dataset in multiple sports scenes. *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 9921–9931.

Luiten, J., Osep, A., Dendorfer, P., Torr, P., Geiger, A., Leal-Taixé, L., & Leibe, B. (2021). HOTA: A higher order metric for evaluating multi-object tracking. *International Journal of Computer Vision*, 129(2), 548–578.